

Morning Talk - Kotlin for Java Developers

What to expect:

Transitioning from Java to Kotlin is less about learning a new language and more about **unlearning** the defensive habits we've built over decades. We start by dismantling the "Billion Dollar Mistake" through Kotlin's native null-safety, ensuring your code fails at compile-time rather than production. You'll see how 50 lines of Java DTOs collapse into a single, readable Kotlin Data Class without losing any functionality.

In the second half, we shift from syntax to strategy. We will map familiar Java patterns (like Streams and Optional) to their idiomatic Kotlin counterparts and explore why **Coroutines** are a more pragmatic choice than raw Threads or Project Loom for high-concurrency tasks - drawing on real-world experience managing **20M+ daily requests**. Finally, we will evaluate the ecosystem, introducing essential libraries like **MockK, Koin, and Arrow**, and compare "Kotlin-first" frameworks like **Ktor** versus the "Kotlin-enhanced" **Spring Boot**.

Talk Agenda:

- **The "Unlearning" Phase:** Dropping semicolons, the `new` keyword, and static overhead.
- **Null Safety in Action:** Moving beyond `if (x != null)` with Safe Calls and Elvis operators.
- **Data Classes & Properties:** Why your IDE's "Generate" menu is about to get very lonely.
- **Smart Casting:** How the compiler does the heavy lifting for your type checks.
- **Feature Mapping:** From `java.util.Stream` to Kotlin **Sequences**.
- **Functional Idioms:** Mastering `let`, `apply`, and `run` to clean up business logic.
- **Concurrency Reimagined:** Coroutines, Structured Concurrency, and the bridge to Project Loom.

- **The Framework Decision:** When to stick with Spring Boot vs. going "Light" with Ktor.
- **Ecosystem Recommendations:** Essential libraries (**MockK, Koin, Arrow**) that every Java dev should know.

Presenter: Aleksei Chursin

Aleksei is a seasoned developer with over 5 years of commercial experience. Currently at **Deutsche Börse**, he develops 24/7 intraday power trading systems managing **+30M daily requests**. With a background in building backend projects from scratch and optimizing high-load systems for merchants like **Nike and Apple** at **Allegro**, Aleksei brings a pragmatic, performance-oriented perspective to Kotlin adoption.